Self-Distilled Agentic Reinforcement Learning: a Literature-Situated Review

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Study repo: `ai-research-studies/self-distilled-agentic-reinforcement-learning`

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Notation

A compact symbol table is reproduced from `02-math-deep-dive.md`; full glossary is in that file.

Symbol	Meaning
$\pi_\theta(\cdot \mid s_t)$	Student policy on student context s_t.
$\pi_T(\cdot \mid s_t^+)$	Teacher branch: same parameters θ, privileged context s_t^+.
y_t	Token sampled by the student at position t.
Δ_t	Teacher–student log-prob gap, $\log \pi_T(y_t \mid s_t^+) - \log \pi_\theta(y_t \mid s_t)$.
h_t	Student entropy, $-\sum_v \pi_\theta(v \mid s_t) \log \pi_\theta(v \mid s_t)$.
$g_t \in [0, 1]$	Per-token sigmoid gate (stop-gradient).
β	Gate sharpness (overloaded with the GRPO KL coefficient).
λ_{SDAR}	Auxiliary loss weight on the SDAR term.

Background: Multi-Turn Agentic Post-Training

Modern post-training of LLM agents is dominated by reinforcement learning with verifiable rewards. Group Relative Policy Optimization (GRPO), introduced by Shao et al. [2024] and scaled by DeepSeek-AI [2025], is now the workhorse: it strips the value head from PPO [Schulman et al., 2017] and replaces the GAE baseline with a group-mean baseline computed across G rollouts of the same prompt. The trajectory-level reward is then broadcast token-wise via a clipped importance ratio. This pipeline is conceptually a descendant of RLHF [Christiano et al., 2017, Ouyang et al., 2022], with the human-preference reward model swapped for a programmatic verifier.

The structural weakness of GRPO in agentic settings is supervisory sparsity. A 20-turn trajectory in ALFWorld [Shridhar et al., 2021] or WebShop [Yao et al., 2022] compresses thousands of token-level decisions into a single scalar reward. *On-Policy Distillation* [Ye et al., 2026, Yang et al., 2026] and its self-distilled variant *OPSD* [Zhao et al., 2026] address this by adding a per-token auxiliary loss against a teacher branch. In the self-distilled regime the teacher is the *same* policy conditioned on a privileged context $s_t^+ = s_t \cup$ (retrieved skill text) available only at training time — a free dense signal. SDAR [Lu et al., 2026] is the agentic extension of this idea, plus a sigmoid gate to handle the multi-turn failure modes documented below.

Method: SDAR

The full objective is

$$\mathcal{L}(\theta) \;=\; \mathcal{L}_{\text{GRPO}}(\theta) \;+\; \lambda_{\text{SDAR}} \cdot \mathcal{L}_{\text{SDAR}}(\theta), \tag{1}$$

with the per-token SDAR loss

$$\ell_t^{\text{SDAR}} \;=\; g_t \cdot \big(\log \pi_T(y_t \mid s_t^+) - \log \pi_\theta(y_t \mid s_t)\big) \;=\; g_t \cdot \Delta_t, \tag{2}$$

aggregated by masked-token average, $\mathcal{L}_{\text{SDAR}} = \big(\sum_t m_t g_t \Delta_t\big)/\big(\sum_t m_t\big)$. The gap Δ_t is exactly the negative single-sample estimator of the per-token reverse KL between teacher and student, evaluated at the student-sampled token — avoiding a full $|\mathcal{V}|$-wide softmax sum.

Lu et al. [2026] benchmark three gating strategies:

$$\begin{aligned}
\text{entropy:}\quad & g_t = \sigma(\beta\, h_t), && (3)\\
\text{gap:}\quad & g_t = \sigma(\beta\, \Delta_t), && (4)\\
\text{soft-OR:}\quad & g_t = \sigma\big(\beta\,[1-(1-h_t)(1-\Delta_t)]\big). && (5)
\end{aligned}$$

Gap gating is the conceptual centerpiece: it operationalises the asymmetric-trust observation. When $\Delta_t > 0$ the privileged context is endorsing the student's sampled token — high-value distillation signal — and the gate opens. When $\Delta_t < 0$ the disagreement is ambiguous (could be skill-retrieval failure rather than a true student error), and the sigmoid attenuates the update. The gate is detached, so the gradient flows only through the student log-prob:

$$\nabla_\theta \mathcal{L}_{\text{SDAR}} \;=\; -\,\text{Agg}\big[\, g_t \cdot \nabla_\theta \log \pi_\theta(y_t \mid s_t)\,\big], \tag{6}$$

which is structurally a gated REINFORCE-style update with g_t as a per-token signed weight.

Results

We summarise Table 1 of Lu et al. [2026] on Qwen 2.5 [Qwen Team, 2024].

Method (Qwen 2.5-3B-Instruct)	ALFWorld ↑	Search-QA ↑	WebShop ↑
GRPO baseline	75.0	36.4	79.8
GRPO + OPSD (ungated)	–	–	–
Skill-GRPO	–	–	–
Skill-SD	–	–	–
RLSD	–	–	–
SDAR (ours)	**84.4**	**43.4**	**85.0**

Numbers reproduced from `02-math-deep-dive.md` §8;
dashes indicate baseline numbers not transcribed into the study notes.

The headline gains are +9.4 pp on ALFWorld average success, +7.0 pp on Search-QA [Dunn et al., 2017], and +5.2 pp on WebShop Score for the 3B model. Qwen 2.5-7B-Instruct shows comparable gains. The qualitative companion result is Figure 2 of Lu et al. [2026]: ungated GRPO+OPSD exhibits catastrophic KL-divergence growth across training, while gated SDAR keeps the per-turn KL bounded — direct empirical support for the multi-turn-instability story.

Position in the Literature

RL backbone. SDAR's GRPO term is taken essentially verbatim from Shao et al. [2024]. The PPO surrogate [Schulman et al., 2017] is therefore the deepest underlying optimisation primitive — with the value head removed and the GAE baseline replaced by a group-mean baseline. SDAR contributes *nothing* to the RL backbone itself; its novelty is the auxiliary objective.

Distillation lineage. The auxiliary term descends from On-Policy Distillation [Ye et al., 2026, Yang et al., 2026], which formalised teacher guidance on student-sampled trajectories, and from On-Policy Self-Distillation [Zhao et al., 2026], which collapses teacher and student into a single set of parameters with a privileged-context teacher branch. SDAR's contribution to this lineage is two-fold: (i) the agentic, multi-turn observation that vanilla OPSD diverges, and (ii) the gating remedy.

Curriculum vs gating. A parallel response to multi-turn OPSD instability is the turn-level curriculum of TCOD [Wang et al., 2026], which schedules teacher influence by turn index. SDAR's gating is the "smooth" alternative — the gate is computed per token from the realised teacher–student gap rather than scheduled exogenously by turn. The two designs are not mutually exclusive; combining them is an obvious follow-up.

RLHF foundations. Conceptually the gated distillation term plays a role analogous to the KL-to-reference regulariser in InstructGPT [Ouyang et al., 2022] and the broader RLHF programme [Christiano et al., 2017]: both prevent the policy from drifting too far from a reference distribution. SDAR replaces the static SFT reference with a dynamic privileged-context teacher and gates the constraint per-token rather than treating it as a fixed Lagrangian.

Sibling agentic-RL work. Hegde and NovaSky-AI Contributors [2026] (the "Reinforcing Recursive Language Models" alphaXiv blog from the NovaSky-AI / SkyRL group) attacks the same supervisory-sparsity problem from a different angle: a shared policy is rolled out across a *recursive call tree*, and child rollouts inherit advantages from their parent. Credit assignment is by tree topology rather than by per-token reliability gating. The two designs are complementary; § Discussion sketches a hybrid.

Benchmarks. The agentic suite is standard: ALFWorld [Shridhar et al., 2021], WebShop [Yao et al., 2022], and Search-QA-style tasks [Dunn et al., 2017]. None of these were designed to stress 50+-turn horizons, which is one reason the depth-scaling question (below) remains open.

Discussion

What the experiments support. Three claims are well-supported by Table 1 and Figure 2 of Lu et al. [2026]: (a) ungated OPSD on top of GRPO destabilises multi-turn training; (b) any of the three proposed gates restores stability; (c) gated SDAR strictly improves over the GRPO baseline on every reported aggregate metric for both 3B and 7B backbones. The KL-bounded behaviour shown in Figure 2 is direct evidence for the failure-mode analysis in § 1 of the paper.

What is hand-waved. The bias–variance argument for gating is informal in the paper — there is no formal theorem that gap-gated OPSD is a lower-variance estimator than ungated OPSD, only an empirical demonstration. Sensitivity to the two new hyperparameters (β and λ_{SDAR}) is not ablated in the main text. The interpretation of Δ_t as a "trust" signal also presupposes that teacher log-prob is a reliable proxy for skill-retrieval quality — a confidently wrong teacher (high Δ_t, low actual quality) would be heavily distilled into the student. This is plausible but not directly tested; the slight under-performance of SDAR vs GRPO+OPSD on some Search-QA sub-tasks is consistent with this critique.

Three future directions. (1) *Depth scaling:* re-run on a benchmark with 50–100 effective turns and check whether the per-turn KL stays bounded. (2) *Adaptive gates:* learn β jointly with θ, or replace the hand-designed gate with $g_\phi(s_t, \Delta_t, h_t)$. (3) *Recursive-call extension:* lift gating into the recursive setting of Hegde and NovaSky-AI Contributors [2026] — gate child rollouts by whether the parent endorses them, combining SDAR-style per-token gating with RLM-style tree-level credit assignment.

See Also

This paper sits inside a broader study chain. The mathematical prerequisites are derived from scratch in Chapter 25 (causal LM as MDP), Chapter 27 (tiny GPT pre-training), Chapter 28 (SFT, RLHF, DPO), Chapter 30 (max-entropy RL), and Chapter 31 (PG / GRPO derivation). The sibling agentic-RL study lives at `pleyva2004/reinforcing-recursive-language-models` — it tackles the same supervisory-sparsity problem with a different mechanism (shared-policy + advantage inheritance across a recursive call tree, vs SDAR's teacher branch + gated per-token distillation), and a hybrid is sketched above.

Reproduction

A locally-verified empirical reproduction of three paper claims (SDAR vs GRPO reward / KL on the toy MDP; entropy-gating KL; ungated GRPO+OPSD collapse) and two measurements of proposed improvements (λ_{SDAR} sensitivity sweep; adaptive-quantile gating tradeoff) is recorded at `findings.md` in the study repository. The findings table includes provenance (machine, numpy version, seed) and per-finding reproduction commands.

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